

Lidar Introduction and Use

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1. Course Content

- Understand the basic parameters of Lidar.
- Start the sample program to view the visualized point cloud of the Lidar.

1. Lidar Introduction

1.1 T-mini Plus Lidar

The ROSMASTER-M3 uses the T-mini Plus Lidar, which is a 360° 2D Lidar. The T-mini Plus is based on the pulsed Time-of-Flight (ToF) ranging principle, combined with relevant optical, electrical, and algorithmic designs to achieve high-frequency and high-precision distance measurement. At the same time, its mechanical structure rotates 360 degrees to continuously acquire angle information, thus achieving 360-degree scanning and ranging, and outputting point cloud data of the scanned environment.

1.2 Lidar Features

- 360-degree omnidirectional scanning, 6-12Hz adjustable scanning frequency
- High-speed ranging, with a ranging frequency of 4000Hz
- Small ranging error and good ranging stability
- Strong resistance to ambient light interference
- Class I eye safety standard

1.3 Performance Parameters

Item	Value	Unit
Ranging Frequency	4000	Hz
Scanning Frequency	6 (6-12)	Hz
Ranging Range	0.05-12	m
Scanning Angle	0-360	Deg
Ranging Accuracy	20	mm
Angular Resolution	0.54	Deg
Pitch Angle	0-1.5	Deg

2. Start the Agent

Note: If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```

microROS_Agent
[1764932423.553407] info | ProxyClient.cpp | create_datareader | d
subscriber created | client_key: 0x14F22590, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1764932423.557165] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x14F22590, topic_id: 0x006(2), participant_
id: 0x000(1)
[1764932423.560435] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x14F22590, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1764932423.565261] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x14F22590, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1764932423.569631] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x14F22590, topic_id: 0x007(2), participant_
id: 0x000(1)
[1764932423.573464] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x14F22590, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1764932423.578415] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x14F22590, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)

```

3. View Lidar Data

- The ROSMASTER-M3 uses a self-developed ROS control board by Yahboom to drive the Lidar.
- After starting the `micro-ROS` agent, the data communication node with the chassis will be launched.
- This node forwards the chassis's data topics. For the Lidar topics:

- Single Lidar version: The published Lidar topics are `/scan2` and `/scan3`. Here, the complete data of a single Lidar is split into left-half and right-half circles and published as two topics.
- Dual Lidar version: The Lidar at the left rear of the car is `/scan0`, and the Lidar at the right front is `/scan1`.
- Here, we take the dual Lidar version as an example to view the data of the `/scan0` topic:

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

Enter the command,

```
ros2 topic echo /scan0
```

```
header:
  stamp:
    sec: 1751267806
    nanosec: 256000000
    frame_id: laser1_frame
  angle_min: 0.0
  angle_max: 6.2831854820251465
  angle_increment: -0.009442231617867947
  time_increment: 0.0
  scan_time: 0.0
  range_min: 0.05000000074505806
  range_max: 12.0
  ranges:
    - 0.7379999756813049
    - 0.746999979019165
    - 0.7329999804496765
    - 0.7170000076293945
    - 0.6809999942779541
    - 0.6110000014305115
```

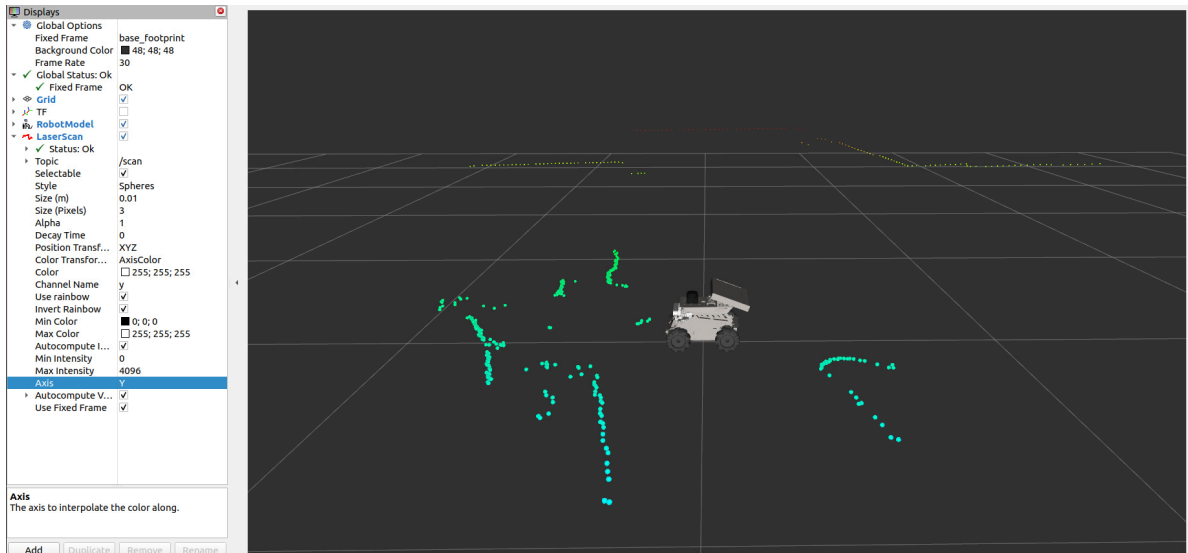
A frame of data is shown in the figure above, where:

- `frame_id`: Represents the name of the Lidar's coordinate system.
- `angle_min` and `angle_max`: Represent the minimum and maximum radians of the Lidar scan, which translates to 0 to 360 degrees.
- `angle_increment`: Represents the angular increment.
- `range_min` and `range_max`: Represent the minimum and maximum scanning distances of the Lidar, in meters. Here, the minimum distance is 0.05 meters, and the maximum is 12.0 meters.
- `ranges`: Represents the distance scanned at each angle within the Lidar's scanning range, in meters.

3. Visualize Lidar Point Cloud

- Use RVIZ to visualize the Lidar's point cloud data.

```
ros2 launch m3_bringup laser_view.launch.py
```



- After starting RVIZ, as shown in the figure above, the colored point cloud is the Lidar's point cloud. By default, it displays the point cloud of the fused `/scan` topic.

[!NOTE]

- For the single Lidar version: Because the Lidar data is transmitted from the chassis as two topics (left-half and right-half circles), the point cloud fusion combines the left-half and right-half data into a complete circle.
 - For the dual Lidar version: The point cloud fusion combines the data from the left-front and right-rear Lidars.
- If you need to view the topic of a single Lidar (dual Lidar version) or a half-circle Lidar (single Lidar version):
 - Click on **Topic** in the **LaserScan** item on the left and select the corresponding topic name to view the corresponding point cloud.

